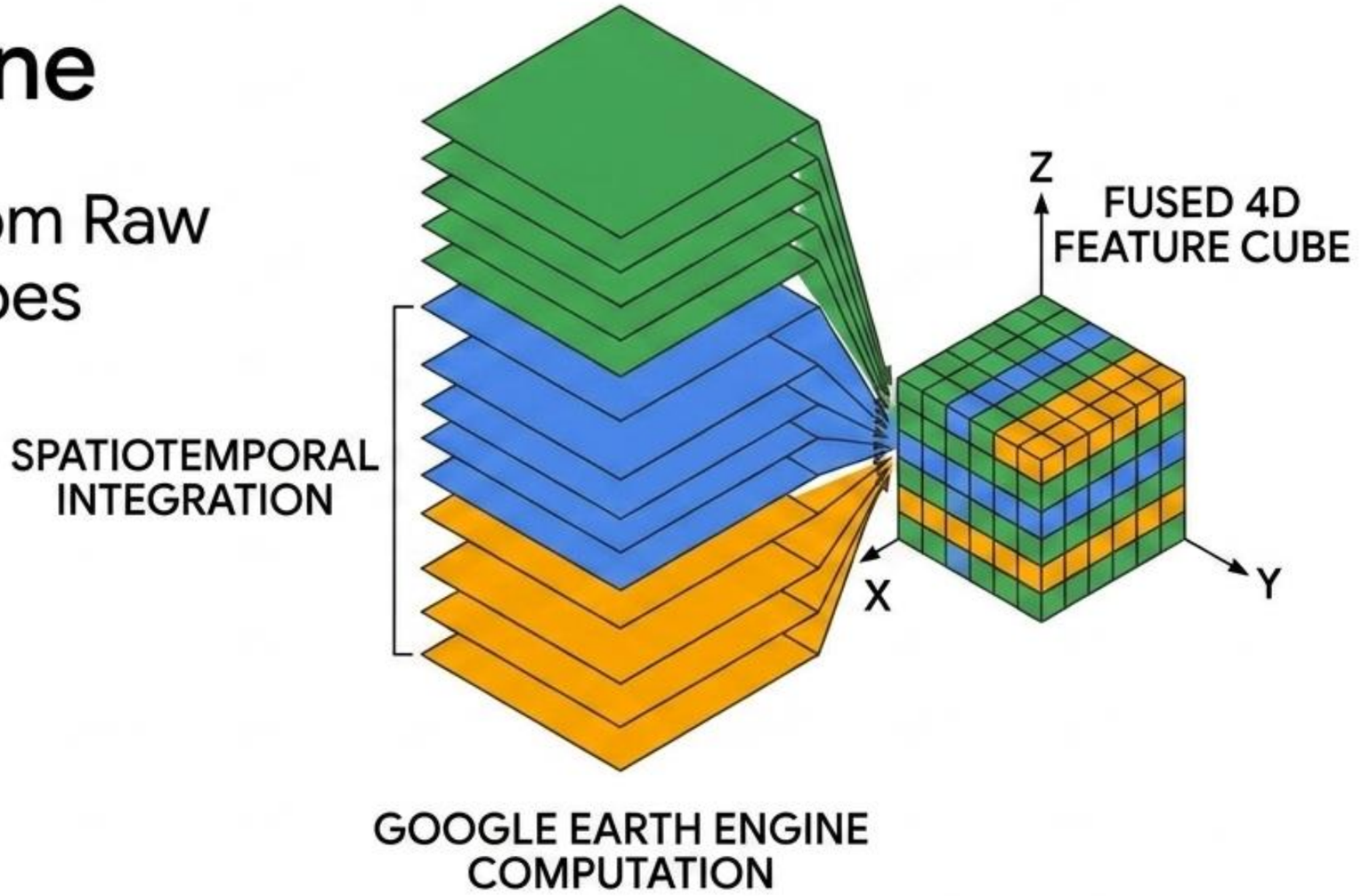
ASEAN
GEOAI FUSION 2026

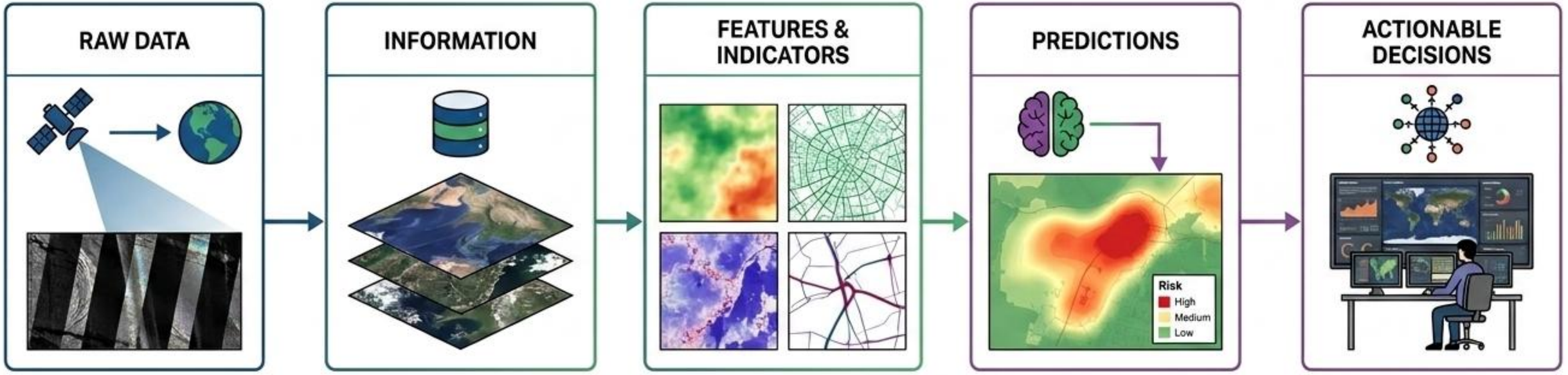
Advanced Data Fusion in Google Earth Engine

Architectural Principles from Raw Sensors to 4D Feature Cubes

Dr. Fathoni Usman, M.Eng., Eng.Tech., IPM, ASEAN Eng.
UNITEN



The GeoAI Objective: Transforming Raw Data into Actionable Intelligence



Raw Data

High-volume, unstructured satellite observations (e.g., raw Sentinel-1/2 swaths) lacking context.

Information (Harmonization)

Cleaned, georeferenced, and aligned datasets ready for analysis.

Features (Extraction)

Derived indices and indicators mapped onto a common grid (AI-ready inputs).

Predictions

Model outputs translating multi-dimensional features into spatial risk and susceptibility.

Intelligence

Actionable mitigation strategies, policy planning, and operational decisions.

The Unseen Gaps: Why Single-Sensor Analysis Fails

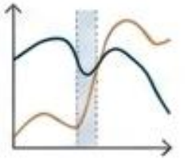
Optical Blind Spots

Optical imagery cannot penetrate cloud cover or heavy smoke. Crucial surface details are frequently obscured in tropical or storm-prone regions.



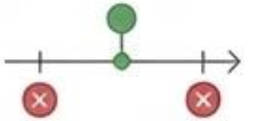
Spectral Ambiguity

A single spectral band range cannot always separate distinct materials that share identical signatures in that specific range but differ elsewhere.



Temporal Gaps

A single satellite overpass misses events that occurred days prior or subsequent, failing to capture dynamic changes.



INSIGHT

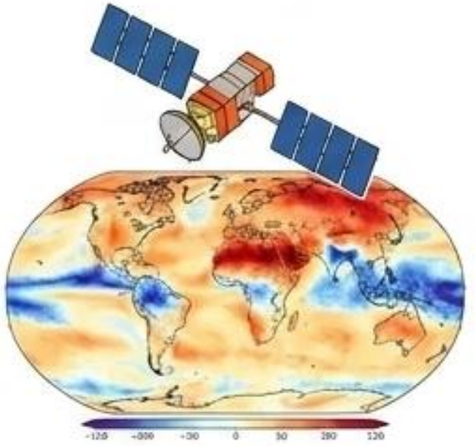
Treating one image or one dataset as the complete truth inherits all of these gaps. Data fusion combines different sources so that what is unseen in one dataset is filled by what another captures well.

The Multi-Sensor Earth Observation Matrix









Optical e.g., Sentinel-2, Landsat	**SAR** e.g., Sentinel-1, ALOS	**Terrain / DEM** e.g., Copernicus, SRTM	**Climate** e.g., ERA5, CHIRPS
High-resolution spectral detail. Ideal for land cover, vegetation indices, and surface water. *Limitation:* Weather-dependent.	Active microwave sensing. All-weather, day/night capability. Penetrates clouds to capture surface structure, roughness, and moisture.	3D structure of the Earth's surface. Provides essential elevation, slope, and drainage context for hazard modelling.	Environmental context. Weather variables (rainfall, temperature anomalies) that drive dynamic dynamic geospatial changes.

Data Harmonization: The Prerequisite for Multi-Sensor Integration

The Alignment Rules

Consistent Scales

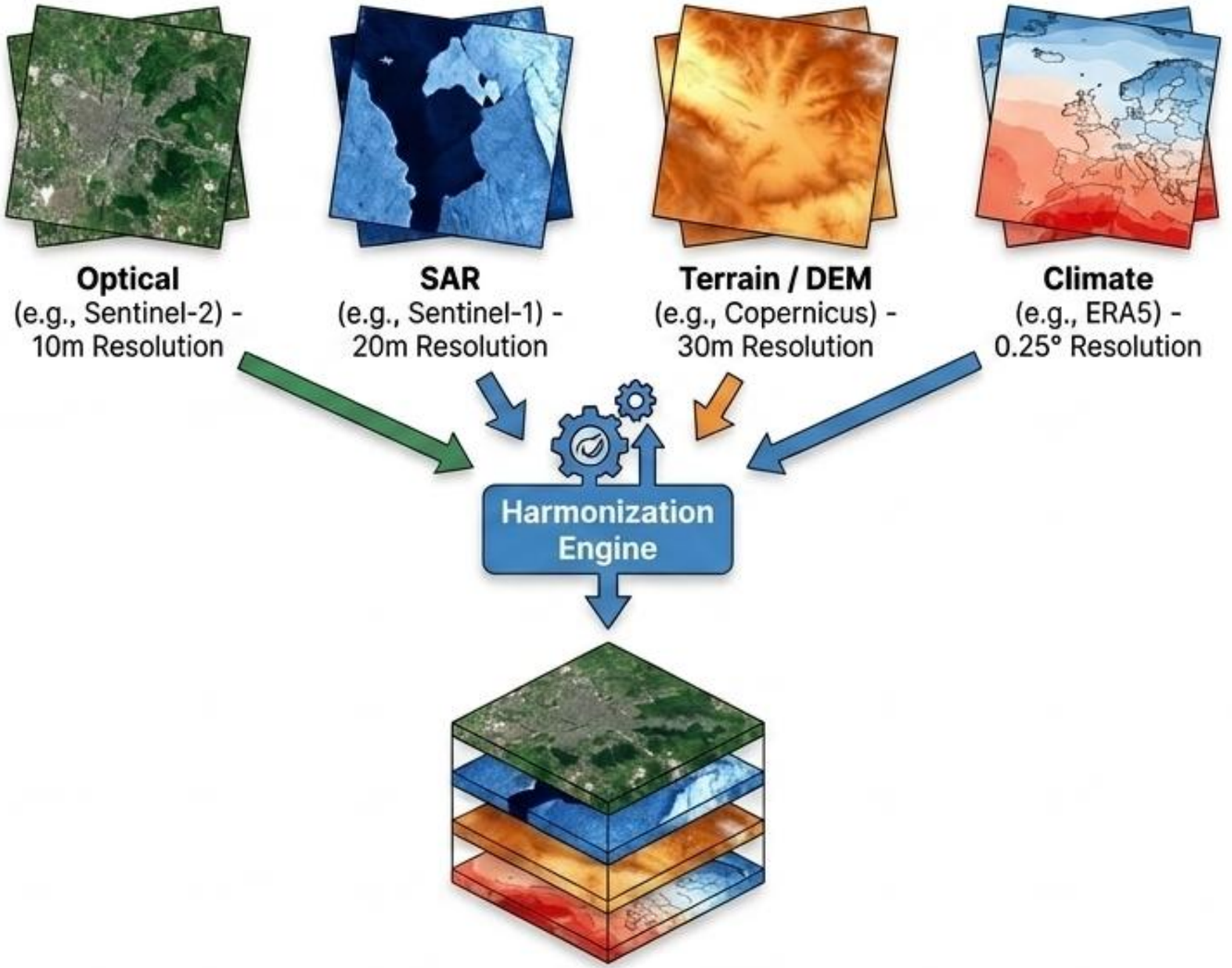
Resampling datasets to match a target spatial resolution (e.g., aligning 20m SAR to 10m Optical).

Consistent Projections

Reprojecting all datasets to a common Coordinate Reference System (CRS) for accurate spatial overlay.

Temporal Alignment

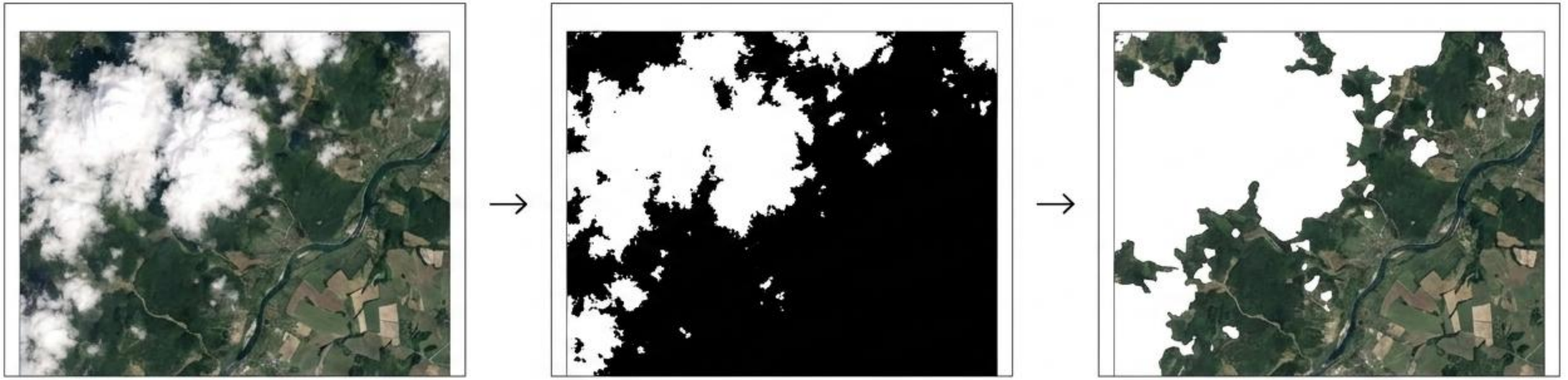
Aggregating observations to match exact time frames (e.g., standardizing daily climate data to match monthly composites).



Unified Multi-Layer Data Cube:
Same Resolution, Same Projection, Same Time Frame, Same Extent.

Pre-Processing: Isolating Clean Observations with QA Masks

CLOUD MASKING WORKFLOW →



1. ORIGINAL IMAGE (Cloudy)

2. CLOUD DETECTION (Using QA Bands)

3. CLOUD MASKED IMAGE
(Clear Observations)

The Contamination Problem

Clouds and cloud shadows fundamentally alter spectral values and corrupt derived indices. You cannot composite contaminated data.

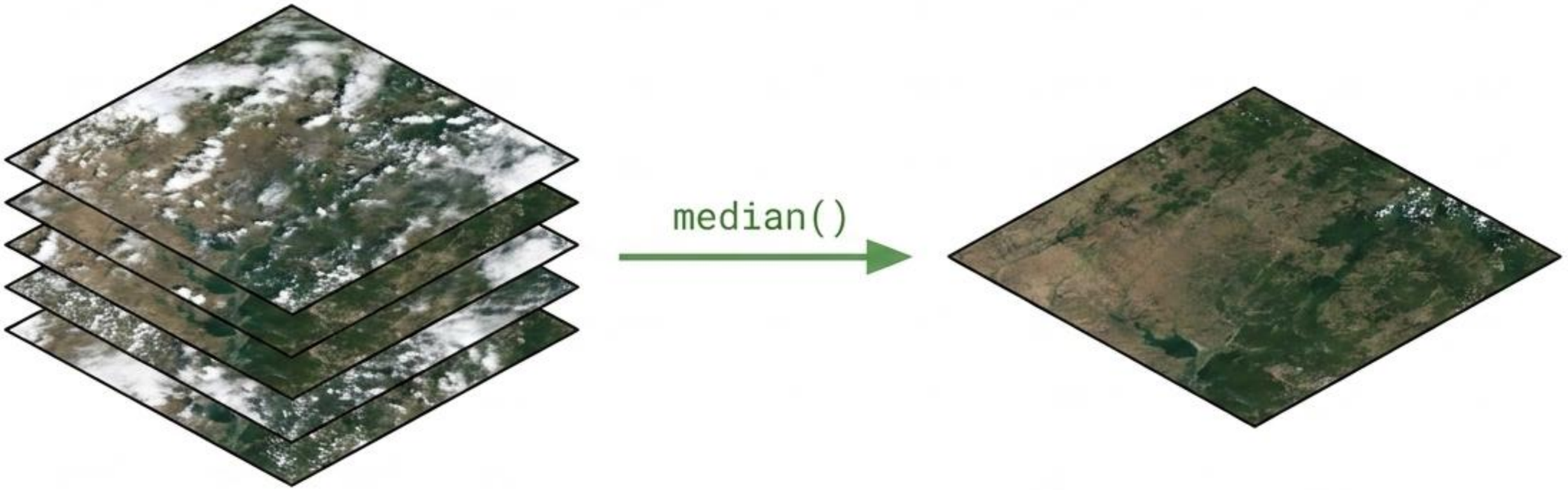
QA Band Thresholding

Sensors provide dedicated Quality Assessment (QA) bands containing pixel-level flags for clouds, cirrus, shadows, and snow/ice.

The Execution

Using bitwise operations to detect and mask (remove) invalid pixels before any statistical reduction takes place.

Compositing: Trading Temporal Detail for Spatial Clarity



Sparse, Daily Observations

The Clean Composite

The Concept

Compiate satellite images readed mask of clouds andmask.

The Mechanic

The Mechanic pormrimate convert median and cleam abuse.

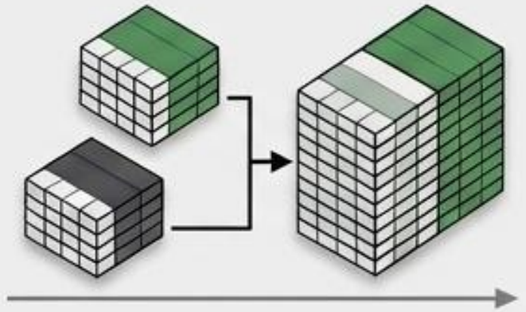
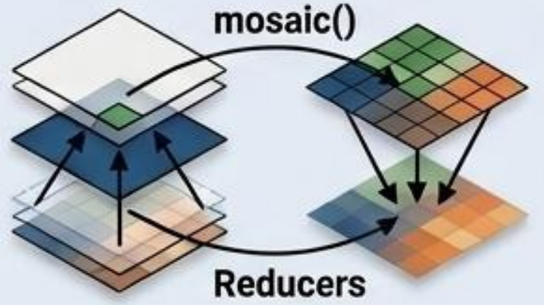
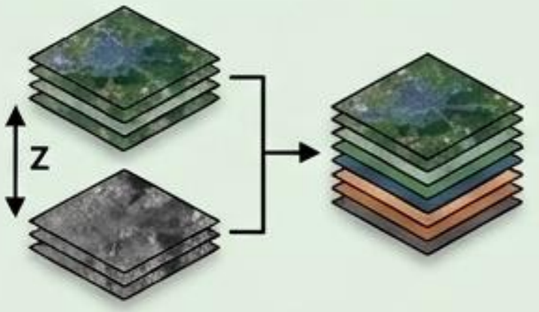
The Benefit

Timeline/and mask of convents to deningstuzlat snave composite.

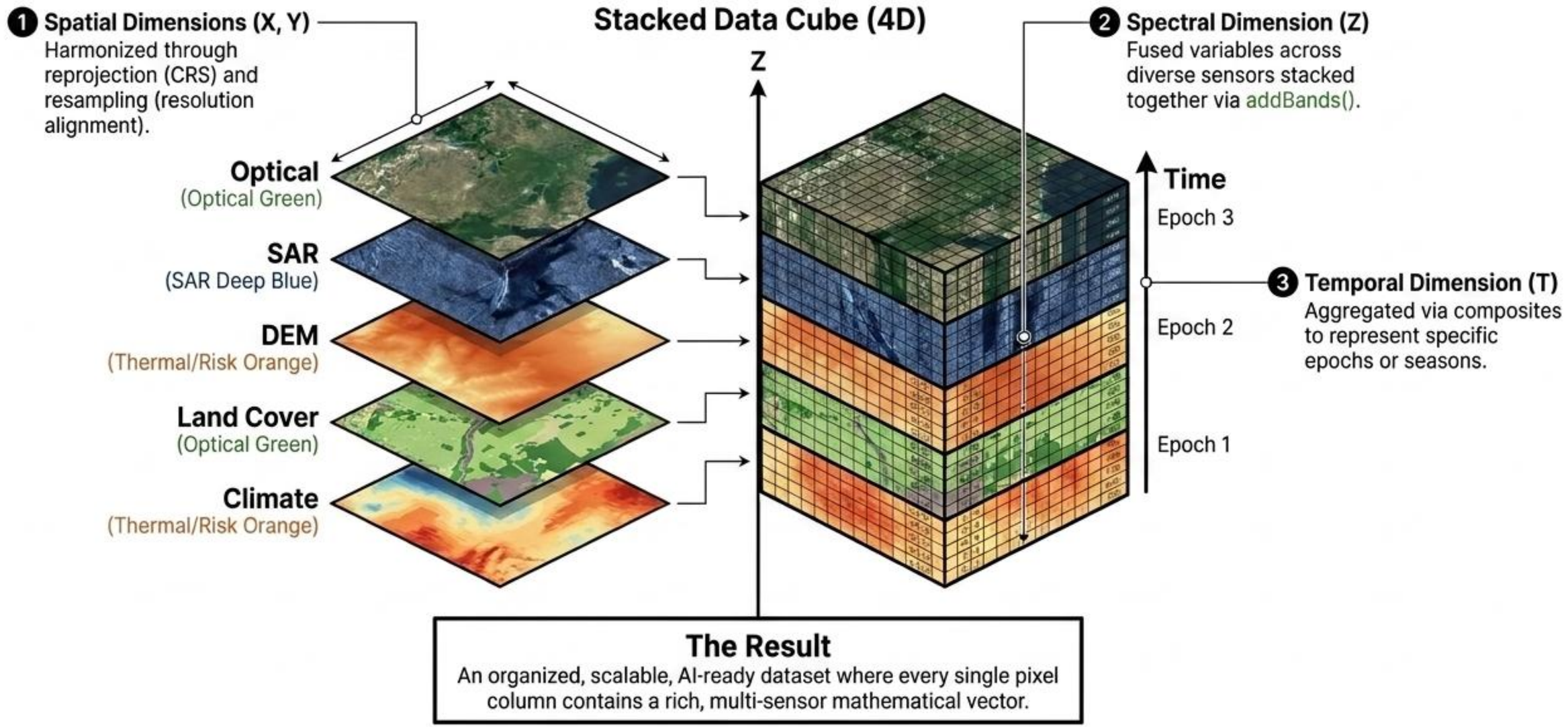
Resampling Matrix: Aligning Disparate Pixel Grids

METHOD	VISUAL EXAMPLE & DESCRIPTION		CRUCIAL RULE
Nearest Neighbor (Default)		Assigns the value of the closest input pixel. Changes no data values.	**Crucial for Categorical Data** (e.g., Land cover classes). Averaging class codes produces meaningless numbers.
Bilinear Interpolation		Uses four neighboring pixels to calculate a weighted average. Produces a smoother output.	**Crucial for Continuous Data** (e.g., Environmental variables, optical bands).
Bicubic Interpolation		Extends the averaging window to sixteen pixels fitting a smooth curve. Highest visual quality.	**Best for visualization of continuous data, but computationally expensive.**

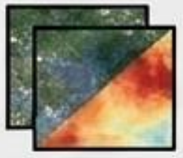
The GEE Merging Toolkit: Connecting Image Geometries

merge() for Collections Concatenates two ImageCollections into one along the temporal axis (e.g., combining Landsat (e.g., combining Landsat 8 and 9).		**Rule:** Collections must share comparable band structures and units.
mosaic() & Reducers for Overlaps Resolves overlapping pixels in the same space. **mosaic()** quickly keeps the top pixel, while reducers like **median()** blend overlapping pixels statistically.		**Rule:** Use **mosaic** for adjacent scenes, use reducers for time-series compositing.
addBands() for Co-located Layers The core operation of data fusion. Stacks bands from a second image onto a first along the Z-axis (e.g., attaching a radar band to an optical image).		**Rule:** Both images must share the **exact same projection** and **pixel grid** first.

The Synthesis: Constructing the 4D Feature Cube

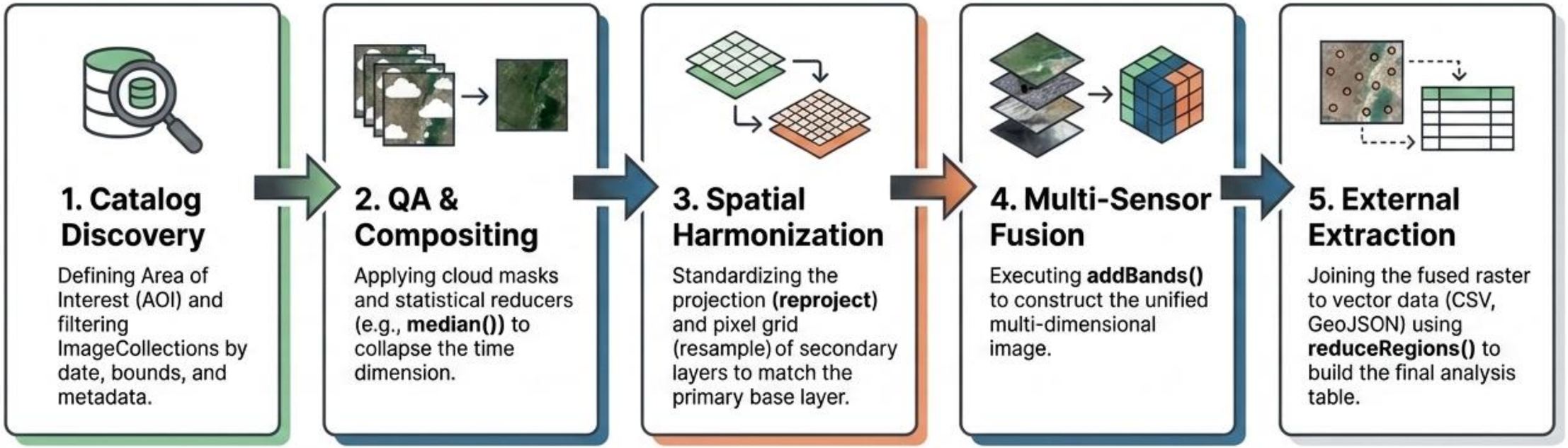


Multi-Sensor Fusion Frameworks in Practice

FUSION FORMULA	ANALYTICAL CAPABILITY
 **Optical** (Green Band) +  **SAR** (Deep Blue Band) →  **Optical + SAR Fusion**	Combines spectral richness (Optical) with all-weather surface structure (SAR). Drastically improves biomass estimation and land cover mapping in cloudy regions.
 **Optical** (Green Band) +  **DEM** (Thermal Orange) →  **Optical + DEM Fusion**	Adds topographic context (elevation, slope, aspect) to spectral data. Essential for landscape analysis, hydrology, and landslide susceptibility.
 **EO Data** (Multi-spectral) +  **Climate** (Thermal Orange) →  **EO Data + Climate Fusion**	Integrates Earth observations with atmospheric variables (rainfall, temperature). Drives drought, flood, and vegetation stress modeling.

DELIVERING ENHANCED ANALYTICAL CAPABILITY THROUGH SYNERGY.

The GEE Data Fusion Architecture



1. Catalog Discovery

Defining Area of Interest (AOI) and filtering ImageCollections by date, bounds, and metadata.

2. QA & Compositing

Applying cloud masks and statistical reducers (e.g., **median()**) to collapse the time dimension.

3. Spatial Harmonization

Standardizing the projection **(reproject)** and pixel grid (resample) of secondary layers to match the primary base layer.

4. Multi-Sensor Fusion

Executing **addBands()** to construct the unified multi-dimensional image.

5. External Extraction

Joining the fused raster to vector data (CSV, GeoJSON) using **reduceRegions()** to build the final analysis table.

Executing the Pipeline: Selection & Compositing

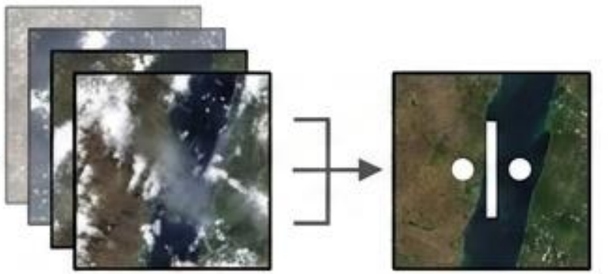
```
// 1. Select and filter datasets
var aoi = ee.Geometry.Rectangle([106.75, -6.30, 106.95, -6.10]);
var s2 = ee.ImageCollection('COPERNICUS/S2_SR_HARMONIZED')
    .filterBounds(aoi).filterDate('2025-06-01', '2025-08-31')
    .filter(ee.Filter.lt('CLOUDY_PIXEL_PERCENTAGE', 20));
var s1 = ee.ImageCollection('COPERNICUS/S1_GRD')
    .filterBounds(aoi).filterDate('2025-06-01', '2025-08-31')
    .filter(ee.Filter.eq('instrumentMode', 'IW'));

// 2. Composite each collection
var s2Composite = s2.median().clip(aoi);
var s1Composite = s1.select(['VV', 'VH']).median().clip(aoi);
```

ANALYTICAL WORKFLOW ANNOTATIONS



Narrows the massive Earth Engine catalog to a specific bounding box and time window.



Statistically collapses dozens of overlapping time-series scenes into a single, clean `s1Composite` and `s2Composite` image.

Executing the Pipeline: Alignment & Data Stacking

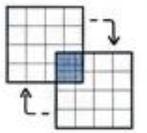
```
1
2  // 3. Harmonize (Resample & Reproject)
3  var s1Aligned = s1Composite
4     .resample('bilinear')
5     .reproject({crs: s2Composite.select('B4').projection()});
6
7  // 4. Fuse layers into a Feature Cube
8  var fused = s2Composite.addBands(s1Aligned);
9  Map.addLayer(fused);
```

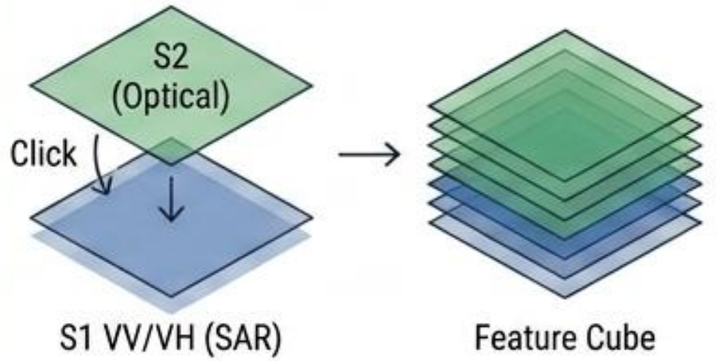


SAR data is continuous; bilinear interpolation ensures a smooth mathematical transition to the Optical grid.



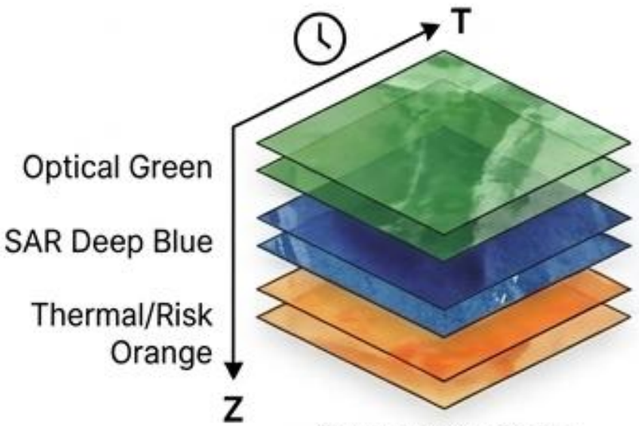
Forces the S1 composite to adopt the exact Coordinate Reference System (CRS) of the S2 band.





Extracting Intelligence: Moving from Raster to Analysis-Ready Table

```
// 5. Combine with External Vector Data
var fieldPoints = ee.FeatureCollection(...);
var extractedData = fused.reduceRegions({
  collection: fieldPoints,
  reducer: ee.Reducer.first(),
  scale: 10
});
```



4D Feature Cube



Ground Truth Points

Pins (GeoJSON, CSV)

The Spatial Join (reduceRegions)

CSV

	Point_ID	Latitude	Longitude	Date	Band1 (Optical)	Band2 (SAA)	indext (NOVI)
1	1	35.6440	-11.9310	12/00/34	0.942	0.453	0.510
2	2	33.5413	-11.3434	12/00/34	0.387	0.375	0.423
3	3	22.6349	-12.3838	13/08/34	0.405	0.575	0.429
4	4	23.6449	-12.3336	12/2K024	0.883	0.761	0.625
5	5	35.5438	-12.9330	12/00/34	0.354	0.653	0.536
0	6	32.5310	-12.3899	12/00/34	0.805	0.646	0.440
7	7	22.6344	-12.3899	13/08/34	0.805	0.665	0.322
6	8	23.6344	-12.5306	12/0K034	0.564	0.906	0.530
9	9	35.6444	-12.9850	12/06/34	0.404	0.657	0.335
10	10	33.5310	-12.9968	12/08/24	0.305	0.687	0.464

Analysis-Ready Table (CSV)

The Spatial Join

reduceRegions() intersects the 4D Feature Cube with ground-truth coordinates (GeoJSON, CSV).

The Output

At every defined point, GEE samples down the Z-axis, extracting optical, SAR, and derived index values simultaneously.

The Final Artifact

The result is no longer an image; it is an analysis-ready tabular dataset—the ultimate foundation for machine learning classifiers and predictive modeling.